TechTriad Hackathon

Introduction

The **TechTriad Hackathon** is a 36-hour challenge focused on reimagining how organizations manage contracts across industries such as BFSI, SaaS, legal services, infrastructure, and procurement.

Despite their critical importance, contracts are often handled through fragmented tools and manual processes, resulting in version inconsistencies, delayed approvals, hidden risks, and disputes due to limited transparency and auditability.

The **TechTriad Hackathon** aims to address these challenges by building a **next-generation Contract Lifecycle Management (CLM) platform**. Participants will develop a unified web-based system that enables organizations to create, review, approve, analyze, and verify contracts securely and efficiently.
The solution **must mandatorily integrate all three technologies**:

- Modern Web UI/UX and frontend workflows
- AI-assisted contract intelligence
- Blockchain-backed trust and auditability

The focus is on building **real-world, enterprise-ready digital tools** where each technology plays a meaningful and necessary role.

Competition Format

1. Preliminary Round (Online)

Participants must submit a **complete system-level proposal** outlining their approach to building the Contract Lifecycle Management platform, based on the three themes listed below. The submission must clearly demonstrate how **Web, AI, and Blockchain components integrate into a single end-to-end application**.

Mandatory components of the submission include:

- Overall system architecture
- UI/UX design and user flows
- Technology stack and integration strategy covering all three themes

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2. Final Round (Onsite at Kshitij)

Selected teams will work on their proposed solution during a **36-hour onsite hackathon** and deliver a working prototype.

TechTriad Hackathon Guidelines

General Guidelines

1. **Deadline Flexibility:** Extensions by one day, if required, will be decided based on progress.
2. **Team Size:**
    - **First Round:** Up to 2 members
    - **Second Round:** Teams of up to 2 members will participate in the 36-hour live

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Submission Requirements

First Round

- **Idea:** Overview of the proposed CLM platform as a single unified product.
- **Approach:** High-level system architecture explaining frontend, backend, AI, and blockchain components and their interactions.
- **Wireframe / Flow Chart:** Mandatory. Must cover contract creation, review and approval flows, AI insights, and audit/verification views.

Second Round

- **GitHub Link:** Proper commits and repository structure.
- **Deployment Link:** Working deployment accessible to judges.

Code Quality

- Proper folder and file structure
- Consistent naming conventions
- Modular and reusable code
- Clear separation of frontend, backend, AI services, and blockchain logic

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Problem Statement – Hackathon Themes

All themes below are mandatory and must be implemented as part of one integrated application.

Theme 1: Contract Management Workbench (UI/UX and Frontend)

Objective

Design and implement a modern, intuitive **Contract Management Workbench** that enables business users, legal teams, and auditors to manage contracts efficiently throughout their lifecycle.

Tasks

Participants must design and implement the following workflows using Next.js / React.js (or an equivalent modern frontend framework):

- **Contract Dashboard**
  - List of contracts with filters (status, type, owner, expiry date)
  - Key contract metrics and alerts surfaced from AI and blockchain layers
- **Contract Creation and Upload**
  - Upload new contracts or create contracts from templates
  - Version management and history tracking
- **Review and Approval Workflow**
  - Multi-step approval flows
  - Role-based access (creator, reviewer, approver)
  - Commenting and annotations linked to contract sections
- **Contract Comparison View**
  - Visual comparison between contract versions
  - Highlight AI-identified clause changes

- **Audit and History View**
  - Unified timeline combining edits, approvals, and blockchain-verified actions

Evaluation Criteria

- Design creativity and user experience
- Implementation of responsive and efficient frontend code
- Clarity of workflows and information architecture

Theme 2: AI Integration – Contract Intelligence

Objective

Integrate AI capabilities that assist users in understanding, reviewing, and managing contracts while remaining accurate, explainable, and reliable.

Tasks

- **Contract Summarization**

  Generate concise summaries highlighting key terms, obligations, and risks
- **Clause Extraction**

  Identify and structure important clauses such as payment terms, termination,liability, and confidentiality
- **Risk Detection**

  1. Flag risky, unusual, or missing clauses.
  2. Provide explanations for identified risks
- **Contract Q&A**

  1. Implement context-aware Q&A grounded strictly in the uploaded contract
  2. Answers must reference the source content

Evaluation Criteria

- Accuracy and usefulness of AI outputs
- Interpretability of summaries and risk flags
- Robust handling of real-world contract text

Theme 3: Blockchain Integration – Trust and Auditability

Objective

Ensure tamper-proof contract integrity and auditability using blockchain as a trust layer, without exposing sensitive contract data.

Tasks

- **Contract Integrity Verification**
  - Store cryptographic hashes of contract versions on-chain
  - Enable verification that a contract has not been altered
- **Approval and Signature Records**
  - Record approval events and signatures immutably
- **Audit Trail**
  - Provide a verifiable history of:
    1. Who approved which version
    2. When changes were made
    3. Which version was finalized
- **Public Verification (Optional)**
  - Allow third-party verification using contract hashes without accessing contract contents

Evaluation Criteria

- Correct and secure blockchain implementation
- Clear justification of on-chain vs off-chain data
- Reliability and usability of audit trails

Judging Criteria

- **Feasibility and Applicability of the Tool:** 35%
- **Innovation and Technical Execution:** 30%
- **User Interface and Experience:** 20%
- **Presentation and Clarity:** 15%

Hackathon Rounds

First Round

- Submit the **idea**, **approach**, and **wireframes / flow charts**.
- Focus on clarity of concept, completeness of system design, and feasibility of the proposed solution.
- Submissions must clearly demonstrate integration of **Web, AI, and Blockchain** components.

Abstract Submission: https://kodeinkgp.in/tech-triad

Second Round (36 Hours)

- Develop a **working prototype** of the proposed platform.
- Submit a **GitHub link** with proper commits and pull requests.
- Provide a **working deployment link** accessible to judges.

Event Schedule

- **First Round Deadline:** 8th January - 11:59 PM
- **Second Round (Live):** 17–18 January (36 hours)
- **Final Judging and Results Announcement:** Within 15 days of the competition

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Prizes

- **Winners:** Cash prize + Internship / Collaboration Opportunity
- **Runner-up:** Cash prize + Certification

The winners must mail the following information (immediately after the announcement of results) to the KodeinKGP Event's Head.

Subject: TechTriad Hackathon, team id – your position
(**example:** TechTriad Hackathon, XX1234 – 1st position)

- Account Holder's Name
- Account Number
- Bank Name
- Branch IFSC Code